exp no: 4 8 Bit division.

AIM:

To write an assembly language program to implement 8 bit division using 8085 processor.

ALGORITHM:

1) Start the program by loading a register pair with the address of memory location.

2) Move the data to a register.

3) Get the second data and load it into the accumulator.

4) Subtract the two register contents.

5) Increment the value of carry.

6) Check whether repeated subtraction is over.

7) Store the value of quotient and the reminder in memory location.

8) Halt.

PROGRAM:

MNEMONICS	EXPLANATION
START : NOP	it is often used for code alignment.
LDA 8500	Load the accumulator with first ~~member~~ in address 8500.
MOV B, A	Move the data from accumulator to B.

LDA 8501	Load the accumulator with the second number in 8501.
MVI C, 00 H	Move the immediate 00 into C.
LOOP: CMP B	Loop: label for loop CMP B: compare the value in accumulator A with
JC LOOP 1	If the carrying is (A< jump to label Loop 1.
SUB B	subtract the value in register (B) from the A.
INR C	Increment register C by 1.
JMP LOOP	Jump back to the beginning of the loop.
STA 8502	Store the data in accumulator 8502.
MOV A, C	move data C to A.
STA 8503	store data in the accumulator in 8503.
RST	Typically transfer control to a predefined interrup service routine
HLT	Halt.

INPUT

ADDRESS : DATA

8500 2
8501 6

OUTPUT:

ADDRESS	DATA
8502	0
8503	3

O/P ✓

RESULT:

Thus the program was executed ~~exe~~ successfully using 8085 processor simulator.

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

A	03	
BC	00	00
DE	00	05
HL	00	0F
PSW	00	00
PC	42	16
SP	FF	FF
Int-Reg	00	

Flag

S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Convertion

Decimal	Hex
0	0

To Hex   To Dec

I/O Ports

0 − + 00

Update Port Value

Memory

2201 − + 03

Update Memory

Load me at

```
LDA 2200
MOV E,A
MVI D,00
LDA 2201
MOV C,A
LXI H,0000
BACK: DAD D
DCR C
JNZ BACK
SHLD 2202
HLT
```

Data | Stack | KeyPad | **Memory** | I/O Ports

Start 2202 OK

Address (Hex)	Address	Data
089A	2202	15
089B	2203	0
089C	2204	0
089D	2205	0
089E	2206	0
089F	2207	0
08A0	2208	0
08A1	2209	0
08A2	2210	0
08A3	2211	0
08A4	2212	0
08A5	2213	0
08A6	2214	0
08A7	2215	0
08A8	2216	0
08A9	2217	0

Line No	Assembler Message
0	Program assembled successfully

Simulator: Idle